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Highly regioselective three-component palladium-catalyzed synthesis of 5-vinyloxazolidin-2-ones from 2,3-allenyl amines, organic iodides, and carbon dioxide

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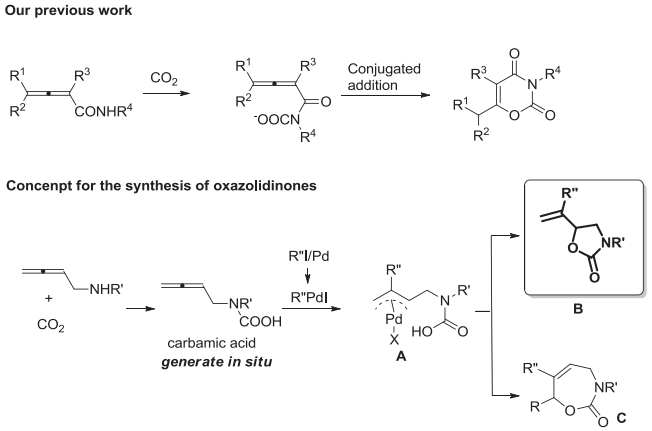
ABSTRACT

A highly regioselective Pd(0)-catalyzed synthesis of 5-vinyloxazolidin-2-ones from 2,3-allenic amines, organic iodides, and CO_2 under mild conditions has been developed.

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1. Introduction

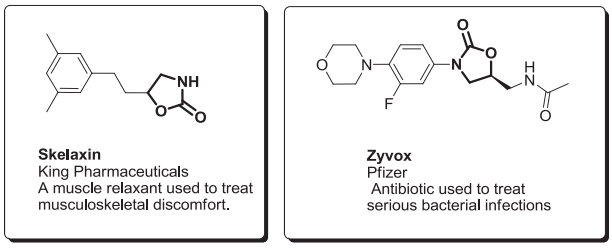
During the last 10 years, more and more attention has been paid to the study on CO_2-activation for the environmental reasons and its abundant, inexpensive, nontoxic, and environmentally benign feature as a C_1 synthon.[1] Challenges are its thermodynamic and kinetic stability. However, amines can readily react with CO_2 to form carbamic acid easily (Scheme 1).[2–8] On the other hand, oxazolidin-2-ones are an important class of compounds that are widely used in organic synthesis and pharmaceutical industry (Scheme 2).[9] Recently, we developed an efficient method for the synthesis of 1,3-oxazine-2,4-diones from CO_2 and 2,3-allenamides (Scheme 1).[8a] On the basis of this work, we envisioned that 5-vinyloxazolidin-2-one **2** may be accessed from carbamic acid in situ generated from 2,3-allenic amine **1** and CO_2 in the presence of palladium catalyst and organic iodides (Scheme 1). However, there may be a challenge in this transformation: both five- and seven-membered ring may be formed via carboxylic anion attack at the π-allylic palladium moiety. Herein, we would like to report a highly regioselective synthesis of 5-vinyloxazolidin-2-ones via a palladium-catalyzed cyclization of 2,3-allenic amines with organic iodides and CO_2 under mild conditions.



Scheme 1. Previous work and our concept for the synthesis of oxazolidinones **B**.

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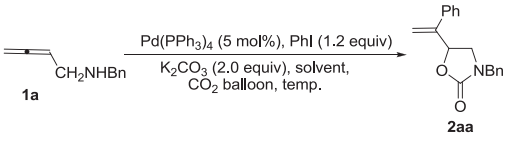


Scheme 2. Best-selling drugs bearing the oxazolidin-2-one substructure.

2. Results and discussion

Initially, 2,3-butadienyl benzyl amine **1a** was chosen to test the feasibility of our hypothesis. Gratifyingly, our expected product 5-vinyloxazolidin-2-one **2aa** was obtained in 77% NMR yield by using 5 mol % of $Pd(PPh_3)_4$, 1.2 equiv of PhI, and 2.0 equiv of K_2CO_3 in DMF at 80 °C in the presence of a CO_2 balloon (Table 1, entry 1). Encouraged by this result, optimization on the reaction conditions was carried out subsequently. As shown in Table 1, DMSO is superior to other solvents such as THF, toluene, and CH_3CN (entries 5–8) while comparable to DMF (entries 2 and 5). Reducing the base or catalyst loading led to a decrease in the NMR yield of **2aa** (entries 9 and 10). Interestingly, a 45% yield of **2aa** was obtained when the reaction was carried out in the absence of external CO_2 (entry 11), where the CO_2 may come from the decomposition of the in situ generated $KHCO_3$ formed from K_2CO_3 and HI. Thus, 2,3-allenic amine reacting with 5 mol % of $Pd(PPh_3)_4$, 1.2 equiv of PhI, and 2.0 equiv of K_2CO_3 in DMSO with a balloon of CO_2 at 70 °C was established as the standard conditions of this reaction.

Table 1
Optimization of reaction conditions[a]

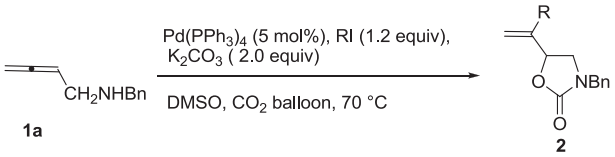


Entry	Solvent	Temp (°C)	Time (h)	Yield of **2aa**[b] (%)
1	DMF	80	8	77
2	DMF	70	9.5	94
3	DMF	60	14.5	92
4	DMF	50	14.5	86
5	DMSO	70	8	96
6	THF	70	8	34[c]
7	Toluene	70	14	40
8	CH_3CN	70	13	62
9[d]	DMSO	70	10	86
10[e]	DMSO	70	10	89
11[f]	DMSO	70	6.7	45

[a] The reaction was carried out with 0.3 mmol of **1a**, 0.36 mmol of phI, 0.6 mmol of K_2CO_3, 0.015 mmol of $Pd(PPh_3)_4$, and a balloon of CO_2 (about 1 L) in 2 mL of the indicated solvent at the indicated temperature.
[b] NMR yield.
[c] Compound **1a** (12%) was recovered.
[d] K_2CO_3 (1 equiv) was used.
[e] $Pd(PPh_3)_4$ (2 mol %) was used.
[f] The reaction was carried out in the absence of the CO_2 balloon.

With the optimized conditions in hand, the scope of the organic iodides was first examined and results are summarized in Table 2. As can be seen in Table 2, different types of organic iodide, including aryl iodides with electron-donating group (entries 2–7) or electron-withdrawing group in the phenyl ring (entries 8–12), heteroaryl iodide (entry 15), and vinyl iodides (entries 16 and 17) are compatible with the reaction. It's noteworthy that aryl iodides with an acidic free OH are also tolerable under the reaction conditions (entries 13 and 14). To check the practicality, this reaction

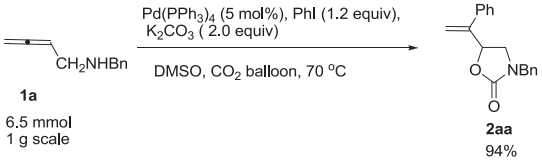
Table 2
The reaction of **1a** with organic iodides[a]



Entry	R	Time (h)	Product	Yield of **2**[b] (%)
1	Ph	8	**2aa**	91
2	*p*-MeC_6H_4	5	**2ab**	92
3	*m*-MeC_6H_4	11	**2ac**	81
4	*p*-$MeOC_6H_4$	4	**2ad**	93
5	*p*-$^{i}PrC_6H_4$	3.5	**2ae**	94
6	3,4-$Me_2C_6H_3$	3	**2af**	82
7	3,4-$(OCH_2CH_2O)C_6H_3$	3.5	**2ag**	84
8	*p*-AcC_6H_4	5	**2ah**	82
9	*p*-FC_6H_4	8	**2ai**	91
10	*p*-$F_3CC_6H_4$	11	**2aj**	85
11	*p*-$EtOOCC_6H_4$	7	**2ak**	86
12	*p*-$NO_2C_6H_4$	4	**2al**	72
13	*p*-HOC_6H_4	3.2	**2am**	73
14	*p*-$HOOCC_6H_4$	11	**2an**	78[c]
15	3-Thienyl	4	**2ao**	94
16	(*E*)-1-Hexenyl	7	**2ap**	82
17	Cinnamenyl	3.5	**2aq**	83

[a] The reaction was carried out with 0.3 mmol of **1a**, 0.36 mmol of RI, 0.6 mmol of K_2CO_3, 0.015 mmol of $Pd(PPh_3)_4$, and a balloon of CO_2 (about 1 L) in 2 mL of DMSO at 70 °C.
[b] Isolated yield after column chromatography unless otherwise indicated.
[c] Yield after column chromatography and recrystallization.

was scaled up to 1 g (6.5 mmol) affording **2aa** in 94% yield [Eq. 1]. The structure of the product was confirmed unambiguously by X-ray diffraction study of **2ab** (Fig. 1).[10]



(1)

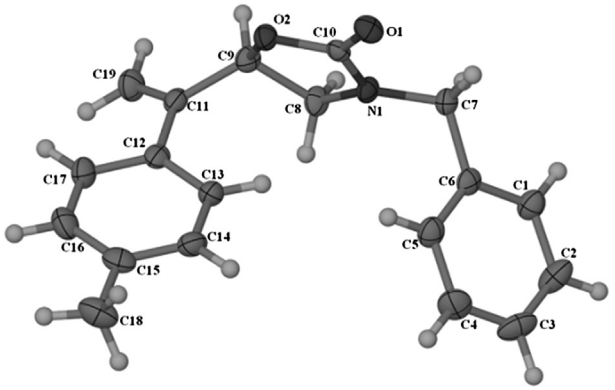
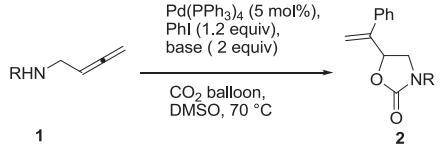


Fig. 1. ORTEP representation of **2ab**.

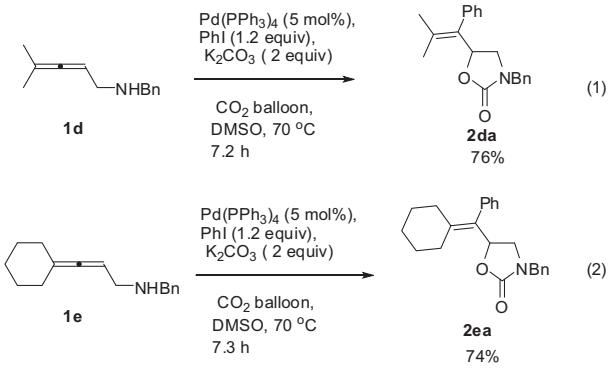
Further study leads to the observation that the basicity of 2,3-allenic amines is crucial to the reaction. The PMB-protected 2,3-allenic amine **1b**, which has a similar basicity with benzyl-protected **1a**, produced **2ba** in 88% yield (Table 3, entry 1). While the less basic phenyl-protected 2,3-allenic amine **1c** was less effective (Table 3, entry 2), thus, a stronger base Cs_2CO_3 was required to afford the product **2ca** in 55% yield (Table 3, entry 3).

Table 3
The effect of N-protecting group in 2,3-allenic amines



Entry	R	Base	Time (h)	Yield of **2** (%)
1	PMB (**1b**)	K_2CO_3	4	88 (**2ba**)
2	Ph (**1c**)	K_2CO_3	9.3	39 (**2ca**)
3	Ph (**1c**)	Cs_2CO_3	4.2	55 (**2ca**)

Next, we turned to investigate the reactivity of substituted 2,3-allenic amines. The reaction **1d** and **1e** went smoothly when 4,4-disubstituted allenic amines were used (Scheme 3).



Scheme 3. Synthesis of 5-vinyloxazolidin-2-one.

3. Conclusion

In conclusion, we have developed a highly efficient Pd(0)-catalyzed synthesis of 5-vinyloxazolidin-2-one from 2,3-allenic amines, organic iodides, and CO_2 under mild conditions with good compatibility of various functional groups. Further studies in this area are being in progress in this laboratory.[11]

4. Experimental section

4.1. General information

All reactions were carried out in oven-dried Schlenk tubes. CH_3CN, DMSO, and DMF were dried over calcium hydride before distillation. THF and toluene were dried over Na wire before distillation. The purity of CO_2 is over 99.995% and used without further purification. All the temperatures are referred to the bath temperatures.

4.2. Synthesis of 5-vinyloxazolidin-2-ones from terminal 2,3-allenic amines or 4,4-disubstituted 2,3-allenic amines, organic iodides, and CO_2

*4.2.1. Synthesis of 3-benzyl-5-(1-phenylvinyl)-2-oxazolidinone (**2aa**). Typical procedure*: To an oven-dried Schlenk tube were added $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (82 mg, 0.593 mmol), DMSO (1 mL), 2,3-allenyl amine **1a** (48 mg, 0.3 mmol), PhI (40 μL, d=1.82 g/mL, 73 mg, 0.36 mmol), and 1 mL of DMSO under nitrogen atmosphere. The mixture was then frozen in a liquid nitrogen bath. After being degassed through one freeze–thaw cycle, the argon inside was completely replaced with CO_2 by using a CO_2 balloon (about 1 L) through a long needle fixed with the rubber stopper. After being stirred at 70 °C (temperature of oil bath) for about 20 min, the needle tip was immersed into the reaction mixture. After being stirred at that temperature for 8 h (total), the reaction was complete as monitored by TLC, and the resulting mixture was diluted with water and extracted with Et_2O. The organic layer was washed with brine, dried over $MgSO_4$, filtrated, and concentrated. The crude product was purified by column chromatography on silica gel (petroleum ether/ethyl acetate=4:1) to afford 77 mg of **2aa** (91%) as a solid: mp 63–64 °C (petroleum ether/ethyl ether); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.28–7.14 (m, 8H, ArH), 7.08 (d, J=7.2 Hz, 2H, ArH), 5.44 (s, 1H, one proton of C=CH_2), 5.37 (s, 1H, one proton of C=CH_2), 5.31 (t, J=8.0 Hz, 1H, OCH), 4.36 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 4.25 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 3.47 (t, J=8.9 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 2.98 (t, J=7.8 Hz, 1H, one proton of NCH_2 in oxazolidinone ring); ^{13}C NMR (75 MHz, $CDCl_3$) δ 157.6, 145.0, 137.0, 135.4, 128.7, 128.6, 128.2, 127.8, 126.5, 114.3, 73.6, 49.0, 48.0; MS (m/z) 279 (M^+, 1.24), 235 (M^+-CO_2, 20.16), 91 (100); IR (KBr, cm^{-1}) ν 1756, 1637, 1492, 1433, 1365, 1330, 1256, 1197, 1101, 1047. Anal. Calcd for $C_{18}H_{17}NO_2$: C 77.40, H 6.13, N 5.01. Found: C 77.42, H 6.26, N 4.82.

The following compounds were prepared according to this procedure:

*4.2.2. 3-Benzyl-5-(1-(p-tolyl)vinyl)-2-oxazolidinone (**2ab**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (85 mg, 0.62 mmol), **1a** (48 mg, 0.30 mmol), and *p*-iodotoluene (79 mg, 0.36 mmol) in 2 mL of DMSO with a balloon of carbon dioxide (about 1 L) afforded 81 mg of **2ab** (92%) as a solid (petroleum ether/ethyl acetate=4:1): mp 85–86 °C (petroleum ether/ethyl ether); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.34–7.24 (m, 3H, ArH), 7.20–7.08 (m, 6H, ArH), 5.47 (s, 1H, one proton of C=CH_2), 5.42–5.33 (m, 2H, one proton of C=CH_2 and OCH), 4.44 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 4.32 (d, J=14.7 Hz, 1H, one proton of CH_2 in Bn), 3.55 (t, J=8.9 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.05 (t, J=7.8 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 2.32 (s, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$) δ 157.7, 144.8, 138.1, 135.4, 134.0, 129.3, 128.6, 127.8, 127.7, 126.3, 113.3, 73.6, 49.1, 48.0, 21.0; MS (m/z) 293 (M^+, 0.51), 249 (M^+-CO_2, 65.57), 91 (100); IR (KBr, cm^{-1}) ν 1737, 1491, 1435, 1321, 1253, 1199, 1104, 1020. Anal. Calcd for $C_{19}H_{19}NO_2$: C 77.79, H 6.53, N 4.77. Found: C 77.68, H 6.76, N 4.75.

*4.2.3. 3-Benzyl-5-(1-(m-tolyl)vinyl)-2-oxazolidinone (**2ac**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (84 mg, 0.61 mmol), **1a** (48 mg, 0.30 mmol), *m*-iodotoluene (78 mg, 0.36 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 72 mg of **2ac** (81%) as an oil (petroleum ether/ethyl acetate=4:1): ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.35–7.00 (m, 9H, ArH), 5.50 (s, 1H, one proton of C=CH_2), 5.43–5.34 (m, 2H, one proton of C=CH_2 and OCH), 4.44 (d, J=14.7 Hz, 1H, one proton of CH_2 in Bn), 4.33 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 3.55 (t, J=8.9 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.06 (t, J=7.7 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 2.32 (s, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$) δ 157.7, 145.2, 138.3, 137.0, 135.5, 129.0, 128.7, 128.5, 127.82, 127.77, 127.3, 123.6, 113.9, 73.6, 49.1, 48.0, 21.3; MS (m/z) 293 (M^+, 1.61), 249 (M^+-CO_2, 56.82), 91 (100); IR (neat, cm^{-1}) ν 1754, 1592, 1491, 1430, 1328, 1255, 1194, 1101, 1043; HRMS calcd for $C_{19}H_{19}NO_2$ (M^+): 293.1416, found 293.1414.

*4.2.4. 3-Benzyl-5-(1-(4-methoxyphenyl)vinyl)-2-oxazolidinone (**2ad**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (83 mg, 0.60 mmol), **1a** (47 mg, 0.30 mmol), 4-iodoanisole (84 mg, 0.36 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 85 mg of **2ad** (93%) as an oil (petroleum ether/ethyl acetate=4:1): ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.35–7.25 (m, 3H, ArH),

7.24–7.10 (m, 4H, ArH), 6.87–6.80 (m, 2H, ArH), 5.43 (s, 1H, one proton of C=CH_2), 5.40–5.32 (m, 2H, one proton of C=CH_2 and OCH), 4.44 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 4.33 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 3.79 (s, 3H, Me), 3.55 (t, J=9.0 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.06 (dd, J_1=8.6 Hz, J_2=7.1 Hz, 1H, one proton of NCH_2 in oxazolidinone ring); ^{13}C NMR (75 MHz, $CDCl_3$) δ 159.6, 157.7, 144.4, 135.4, 129.3, 128.6, 127.80, 127.76, 127.7, 114.0, 112.9, 73.7, 55.1, 49.1, 48.0; MS (m/z) 309 (M^+, 13.96), 265 (M^+-CO_2, 41.47), 91 (100); IR (neat, cm^{-1}) ν 1753, 1607, 1510, 1432, 1251, 1184, 1103, 1031; HRMS calcd for $C_{19}H_{19}NO_3$ (M^+): 309.1365, found 309.1363.

*4.2.5. 3-Benzyl-5-(1-(4-isopropylphenyl)vinyl)-2-oxazolidinone (**2ae**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (83 mg, 0.60 mmol), **1a** (49 mg, 0.30 mmol), 1-iodo-4-isopropylbenzene (88 mg, 0.36 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 93 mg of **2ae** (94%) as an oil (petroleum ether/ethyl acetate=4:1): ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.35–7.24 (m, 3H, ArH), 7.23–7.12 (m, 6H, ArH), 5.47 (s, 1H, one proton of C=CH_2), 5.45–5.34 (m, 2H, one proton of C=CH_2 and OCH), 4.45 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 4.33 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 3.56 (t, J=9.0 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.07 (t, J=7.8 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 2.88 (hept, J=6.9 Hz, 1H, CH in *i*-Pr), 1.23 (d, J=6.6 Hz, 6H, Me in *i*-Pr); ^{13}C NMR (75 MHz, $CDCl_3$) δ 157.7, 149.0, 144.8, 135.4, 134.4, 128.6, 127.8, 127.7, 126.6, 126.4, 113.3, 73.6, 49.2, 48.0, 33.6, 23.7; MS (m/z) 321 (M^+, 1.03), 277 (M^+-CO_2, 74.74), 91 (100); IR (neat, cm^{-1}) ν 1756, 1598, 1490, 1431, 1329, 1254, 1194, 1100, 1048; HRMS calcd for $C_{21}H_{23}NO_2$ (M^+): 321.1729, found 321.1726.

*4.2.6. 3-Benzyl-5-(1-(3,4-dimethylphenyl)vinyl)-2-oxazolidinone (**2af**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (85 mg, 0.62 mmol), **1a** (48 mg, 0.30 mmol), 3,4-dimethyliodobenzene (83 mg, 0.36 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 76 mg of **2af** (82%) as an oil (petroleum ether/ethyl acetate=4:1): ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.34–7.22 (m, 3H, ArH), 7.20–7.13 (m, 2H, ArH), 7.11–6.94 (m, 3H, ArH), 5.46 (s, 1H, one proton of C=CH_2), 5.45–5.33 (2H, one proton of C=CH_2 and OCH), 4.44 (d, J=14.7 Hz, 1H, one proton of CH_2 in Bn), 4.33 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 3.54 (t, J=8.9 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.06 (t, J=7.7 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 2.23 (s, 6H, Me×2); ^{13}C NMR (75 MHz, $CDCl_3$) δ 157.7, 145.0, 136.8, 135.5, 134.6, 129.8, 128.6, 127.8, 127.7, 123.8, 113.0, 73.6, 49.2, 48.0, 19.7, 19.3; MS (m/z) 307 (M^+, 3.27), 263 (M^+-CO_2, 73.05), 91 (100); IR (neat, cm^{-1}) ν 1755, 1600, 1494, 1432, 1329, 1255, 1195, 1100, 1046; HRMS calcd for $C_{20}H_{21}NO_2$ (M^+): 307.1572, found 307.1572.

*4.2.7. 3-Benzyl-5-(1-(2,3-dihydrobenzo[b][1,4]dioxin-6-yl)vinyl)-2-oxazolidinone (**2ag**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (84 mg, 0.61 mmol), **1a** (50 mg, 0.31 mmol), 6-iodo-2,3-dihydrobenzo[1,4]dioxine (92 mg, 0.35 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 89 mg of **2ag** (84%) as an oil (petroleum ether/ethyl acetate=4:1): ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.38–7.24 (m, 3H, ArH), 7.23–7.12 (m, 2H, ArH), 6.85–6.70 (m, 3H, ArH), 5.43 (s, 1H, one proton of C=CH_2), 5.39–5.29 (m, 2H, one proton of C=CH_2 and OCH), 4.46 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 4.33 (d, J=15.3 Hz, 1H, one proton of CH_2 in Bn), 4.22 (s, 4H, OCH_2CH_2O), 3.56 (t, J=8.7 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.06 (t, J=7.5 Hz, 1H, one proton of NCH_2 in oxazolidinone ring); ^{13}C NMR (75 MHz, $CDCl_3$) δ 157.6, 144.2, 143.7, 143.4, 135.4, 130.3, 128.7, 127.80, 127.76, 119.5, 117.3, 115.3, 112.7, 73.4, 64.25, 64.17, 49.2, 48.0; MS (m/z) 337 (M^+, 15.59), 293 (M^+-CO_2, 11.13), 91 (100); IR (neat, cm^{-1}) ν 1753, 1578, 1505, 1431, 1294, 1251, 1194, 1100, 1060; HRMS calcd for $C_{20}H_{19}NO_4$ (M^+): 337.1314, found 337.1314.

*4.2.8. 5-(1-(4-Acetylphenyl)vinyl)-3-benzyl-2-oxazolidinone (**2ah**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (82 mg, 0.59 mmol), **1a** (48 mg, 0.30 mmol), 1-(4-iodophenyl) ethanone (89 mg, 0.36 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 79 mg of **2ah** (82%) as an oil (petroleum ether/ethyl acetate=3:1): ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.91 (d, J=7.5 Hz, 2H, ArH), 7.38 (d, J=7.8 Hz, 2H, ArH), 7.34–7.23 (m, 3H, ArH), 7.22–7.10 (m, 2H, ArH), 5.62 (s, 1H, one proton of C=CH_2), 5.56 (s, 1H, one proton of C=CH_2), 5.42 (t, J=7.8, 1H, OCH), 4.43 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 4.35 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 3.60 (t, J=8.9 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.07 (t, J=7.7 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 2.59 (s, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$) δ 197.3, 157.4, 144.2, 141.6, 136.6, 135.2, 128.7, 128.6, 127.8, 126.8, 116.3, 73.3, 48.8, 48.0, 26.5; MS (m/z) 321 (M^+, 1.94), 277 (M^+-CO_2, 47.91), 91 (100); IR (neat, cm^{-1}) ν 1754, 1682, 1604, 1491, 1431, 1361, 1264, 1195, 1100, 1043; HRMS calcd for $C_{20}H_{19}NO_3$ (M^+): 321.1365, found 321.1363.

*4.2.9. 3-Benzyl-5-(1-(4-fluorophenyl)vinyl)-2-oxazolidinone (**2ai**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (85 mg, 0.62 mmol), **1a** (47 mg, 0.30 mmol), 4-fluoroiodobenzene (80 mg, 0.36 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 80 mg of **2ai** (91%) as an oil (petroleum ether/ethyl acetate=4:1): ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.37–7.20 (m, 5H, ArH), 7.20–7.09 (m, 2H, ArH), 7.00 (t, J=8.3 Hz, 2H, ArH), 5.50 (s, 1H, one proton of C=CH_2), 5.40 (s, 1H, one proton of C=CH_2), 5.35 (t, J=8.0 Hz, 1H, OCH), 4.41 (d, J=15.3 Hz, 1H, one proton of CH_2 in Bn), 4.35 (d, J=15.3 Hz, 1H, one proton of CH_2 in Bn), 3.56 (t, J=8.9 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.06 (t, J=7.7 Hz, 1H, one proton of NCH_2 in oxazolidinone ring); ^{13}C NMR (75 MHz, $CDCl_3$) δ 162.6 (d, J_{C-F}=246.7 Hz), 157.5, 144.1, 135.3, 133.1 (d, J_{C-F}=3.3 Hz), 128.7, 128.5 (d, J_{C-F}=8.2 Hz), 127.8, 115.6 (d, J=21.4 Hz), 115.0, 73.8, 48.8, 48.0; ^{19}F NMR (282 MHz, $CDCl_3$) δ −113.3; MS (m/z) 297 (M^+, 1.53), 253 (M^+-CO_2, 40.02), 91 (100); IR (neat, cm^{-1}) ν 1754, 1601, 1506, 1432, 1329, 1234, 1100, 1043. Anal. Calcd for $C_{18}H_{16}FNO_2$: C 72.71, H 5.42, N 4.71. Found: C 72.63, H 5.62, N 4.58.

*4.2.10. 3-Benzyl-5-(1-(4-(trifluoromethyl)phenyl)vinyl)-2-oxazolidinone (**2aj**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (85 mg, 0.62 mmol), **1a** (47 mg, 0.30 mmol), 4-trifluoromethyliodobenzene (97 mg, 0.36 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 87 mg of **2aj** (85%) as a solid (petroleum ether/ethyl acetate=4:1): mp 72–73 °C (petroleum ether/ethyl ether); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.58 (d, J=8.1 Hz, 2H, ArH), 7.39 (d, J=8.1 Hz, 2H, ArH), 7.34–7.22 (m, 3H, ArH), 7.20–7.08 (m, 2H, ArH), 5.60 (s, 1H, one proton of C=CH_2), 5.51 (s, 1H, one proton of C=CH_2), 5.39 (t, J=8.0, 1H, OCH), 4.38 (s, 2H, CH_2 in Bn), 3.58 (t, J=9.0 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.06 (t, J=7.8 Hz, 1H, one proton of NCH_2 in oxazolidinone ring); ^{13}C NMR (125 MHz, $CDCl_3$) δ 157.5, 144.2, 140.7, 135.3, 130.4 (q, J=32.5 Hz), 128.8, 128.0, 127.9, 127.3, 125.6 (q, J=3.7 Hz), 123.9 (q, J=270.6 Hz), 117.0, 73.7, 48.8, 48.2; ^{19}F NMR (282 MHz, $CDCl_3$) δ −62.7; MS (m/z) 347 (M^+, 10.13), 303 (M^+-CO_2, 8.93), 91 (100); IR (neat, cm^{-1}) ν 1750, 1614, 1487, 1435, 1320, 1252, 1165, 1115, 1057. Anal. Calcd for $C_{19}H_{16}F_3NO_2$: C 65.70, H 4.64, N 4.03. Found: C 65.84, H 4.66, N 4.08.

*4.2.11. 3-Benzyl-5-(1-(4-(ethoxycarbonyl)phenyl)vinyl)-2-oxazolidinone (**2ak**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (84 mg, 0.61 mmol), **1a** (49 mg, 0.31 mmol), ethyl 4-iodobenzoate (98 mg, 0.36 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 93 mg of **2ak**

(86%) as an oil (petroleum ether/ethyl acetate=4:1): ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.99 (d, J=8.4 Hz, 2H, ArH), 7.40–7.24 (m, 5H, ArH), 7.20–7.12 (m, 2H, ArH), 5.60 (s, 1H, one proton of C=CH_2), 5.54 (s, 1H, one proton of C=CH_2), 5.42 (t, J=8.0 Hz, 1H, OCH), 4.46–4.28 (m, 4H, CO_2CH_2 and CH_2 in Bn), 3.60 (t, J=8.9 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.06 (dd, J_1=8.7 Hz, J_2=6.9 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 1.39 (t, J=7.1 Hz, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$) δ 165.8, 157.4, 144.3, 141.3, 135.2, 130.1, 129.7, 128.6, 128.0, 127.7, 126.5, 116.0, 73.3, 60.9, 48.7, 47.9, 14.1; MS (m/z) 351 (M^+, 1.07), 307 (M^+-CO_2, 53.72), 91 (100); IR (neat, cm^{-1}) ν 1757, 1715, 1607, 1490, 1434, 1367, 1277, 1188, 1104, 1017; HRMS calcd for $C_{21}H_{21}NO_4$ (M^+): 351.1471, found 351.1471.

*4.2.12. 3-Benzyl-5-(1-(4-nitrophenyl)vinyl)-2-oxazolidinone (**2al**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (84 mg, 0.60 mmol), **1a** (48 mg, 0.30 mmol), 1-iodo-4-nitrobenzene (91 mg, 0.36 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 70 mg of **2al** (72%) as a solid (petroleum ether/ethyl acetate=3:1): mp 77–78 °C (petroleum ether/ethyl ether); ^{1}H NMR (300 MHz, $CDCl_3$) δ 8.16 (d, J=7.8 Hz, 2H, ArH), 7.45 (d, J=8.1 Hz, 2H, ArH), 7.35–7.25 (m, 3H, ArH), 7.21–7.12 (m, 2H, ArH), 5.68 (s, 1H, one proton of C=CH_2), 5.60 (s, 1H, one proton of C=CH_2), 5.41 (t, J=7.8 Hz, 1H, OCH), 4.39 (s, 2H, CH_2 in Bn), 3.63 (t, J=8.9 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.08 (t, J=7.8 Hz, 1H, one proton of NCH_2 in oxazolidinone ring); ^{13}C NMR (75 MHz, $CDCl_3$) δ 157.2, 147.5, 143.5, 135.1, 128.8, 128.01, 127.95, 127.7, 123.8, 118.3, 73.3, 48.6, 48.1; MS (m/z) 324 (M^+, 2.26), 91 (100); IR (KBr, cm^{-1}) ν 1751, 1635, 1594, 1515, 1439, 1340, 1246, 1193, 1102, 1037, 1004. Anal. Calcd for $C_{18}H_{16}N_2O_4$: C 66.66, H 4.97, N 8.64. Found: C 66.51, H 5.10, N 8.50.

*4.2.13. 3-Benzyl-5-(1-(4-hydroxyphenyl)vinyl)-2-oxazolidinone (**2am**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (83 mg, 0.60 mmol), **1a** (49 mg, 0.31 mmol), 4-iodophenol (79 mg, 0.36 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 66 mg of **2am** (73%) as an oil (petroleum ether/ethyl acetate=2:1): ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.35 (s, 1H, OH), 7.32–7.23 (m, 3H, ArH), 7.15–7.07 (m, 4H, ArH), 6.87–6.79 (m, 2H, ArH), 5.43–5.21 (m, 3H, C=CH_2 and OCH), 4.42 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 4.33 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 3.56 (t, J=9.0 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.10 (dd, J_1=8.6 Hz, J_2=7.1 Hz, 1H, one proton of NCH_2 in oxazolidinone ring); ^{13}C NMR (75 MHz, $CDCl_3$) δ 158.4, 156.7, 144.4, 135.1, 128.8, 128.5, 128.3, 128.0, 127.9, 127.8, 115.7, 113.3, 74.5, 49.1, 48.1; MS (m/z) 295 (M^+, 4.56), 251 (M^+-CO_2, 56.55), 91 (100); IR (neat, cm^{-1}) ν 3298 (br), 1726, 1604, 1508, 1443, 1363, 1332, 1263, 1106, 1040, 1006; HRMS calcd for $C_{18}H_{17}NO_3$ (M^+): 295.1208, found 295.1207.

*4.2.14. 3-Benzyl-5-(1-(4-carboxylphenyl)vinyl)-2-oxazolidinone (**2an**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (85 mg, 0.62 mmol), **1a** (50 mg, 0.31 mmol), 4-iodobenzoic acid (88 mg, 0.35 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO in 70 °C for 11 h was quenched by 1 M HCl, and extracted with EtOAc and the organic layer was washed with brine, dried over $MgSO_4$, filtrated, and concentrated. The crude product was purified by column chromatography on silica gel (CH_2Cl_2/CH_3OH=25:1) to afford 93 mg (slightly impure) of **2an** as a solid, which was further purified by recrystallization (petroleum ether/ethyl acetate) affording 79 mg of pure **2an** (78%): mp 169–170 °C (petroleum ether/ethyl acetate); ^{1}H NMR (300 MHz, CD_3OD) δ 7.96 (d, J=8.1 Hz, 2H, ArH), 7.41 (d, J=8.1 Hz, 2H, ArH), 7.32–7.20 (m, 3H, ArH), 7.18–7.05 (m, 2H, ArH), 5.61–5.50 (m, 3H, C=CH_2 and CH), 4.44 (d, J=15.3 Hz, 1H, one proton of CH_2 in Bn), 4.26 (d, J=15.3 Hz, 1H, one proton of CH_2 in Bn), 3.72 (t, J=9.2 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.08 (t, J=7.7 Hz, 1H, one proton of NCH_2 in oxazolidinone ring); ^{13}C NMR (75 MHz, CD_3OD) δ 169.3, 159.9, 146.7, 143.0, 136.9, 131.8, 131.1, 129.8, 128.9, 128.1, 116.8, 75.5, 49.9; MS (m/z) 323 (M^+, 2.89), 279 (M^+-CO_2, 16.56), 91 (100); IR (KBr, cm^{-1}) ν 3500–2200 (br), 1755, 1686, 1607, 1564, 1495, 1432, 1368, 1326, 1298, 1269, 1191, 1098, 1055, 1035, 1009. Anal. Calcd for $C_{19}H_{17}NO_4$: C 70.58, H 5.30, N 4.33. Found: C 70.33, H 5.31, N 4.33.

*4.2.15. 3-Benzyl-5-(1-(3-thienyl)vinyl)-2-oxazolidinone (**2ao**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (85 mg, 0.62 mmol), **1a** (48 mg, 0.30 mmol), 3-iodothiophene (78 mg, 0.37 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 81 mg of **2ao** (94%) as an oil (petroleum ether/ethyl acetate=4:1): ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.36–7.18 (m, 6H, ArH), 7.00–6.93 (m, 1H, H in thienyl), 6.90 (d, J=3.6 Hz, 1H, H in thienyl), 5.54 (s, 1H, one proton of C=CH_2), 5.42 (s, 1H, one proton of C=CH_2), 5.32 (t, J=8.0 Hz, 1H, OCH), 4.49 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 4.35 (d, J=14.7 Hz, 1H, one proton of CH_2 in Bn), 3.67 (t, J=9.0 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.19 (t, J=7.7 Hz, 1H, one proton of NCH_2 in oxazolidinone ring); ^{13}C NMR (75 MHz, $CDCl_3$) δ 157.4, 139.5, 138.7, 135.3, 128.7, 127.91, 127.87, 127.5, 125.3, 124.0, 112.5, 73.2, 49.4, 48.1; MS (m/z) 285 (M^+, 3.48), 241 (M^+-CO_2, 54.57), 91 (100); IR (neat, cm^{-1}) ν 2951, 2919, 2850, 1748, 1621, 1430, 1374, 1325, 1248, 1190, 1012; HRMS calcd for $C_{16}H_{15}NO_2S$ (M^+): 285.0824, found 285.0823.

*4.2.16. (E)-3-Benzyl-5-(octa-1,3-dien-2-yl)-2-oxazolidinone (**2ap**).* The reaction of $Pd(PPh_3)_4$ (17.3 mg, 0.015 mmol), K_2CO_3 (83.5 mg, 0.61 mmol), **1a** (47 mg, 0.30 mmol), (*E*)-1-iodohex-1-ene (75.7 mg, 0.36 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 69 mg of **2ap** (82%) as an oil (petroleum ether/ethyl acetate=4:1): ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.40–7.20 (m, 5H, ArH), 5.99 (d, J=16.2 Hz, 1H, C–CH=C), 5.54–5.42 (m, 1H, C=CH), 5.27 (s, 1H, one proton of C=CH_2), 5.14 (t, J=8.0 Hz, 1H, OCH), 5.09 (s, 1H, one proton of C=CH_2), 4.49 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 4.37 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 3.65 (t, J=8.9 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.14 (t, J=7.7 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 2.05 (q, J=6.7 Hz, 2H, C=CCH_2), 1.40–1.20 (m, 4H, CH_2CH_2), 0.88 (t, J=6.9 Hz, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$) δ 157.7, 142.9, 135.6, 131.8, 128.7, 127.9, 127.8, 127.7, 112.9, 72.1, 49.6, 48.1, 32.6, 31.0, 22.0, 13.7; MS (m/z) 285 (M^+, 6.06), 241 (M^+-CO_2, 4.13), 91 (100); IR (neat, cm^{-1}) ν 1754, 1650, 1609, 1492, 1430, 1362, 1331, 1253, 1203, 1102, 1055; HRMS calcd for $C_{18}H_{23}NO_2$ (M^+): 285.1729, found 285.1729.

*4.2.17. (E)-3-Benzyl-5-(4-phenylbuta-1,3-dien-2-yl)-2-oxazolidinone (**2aq**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (83 mg, 0.60 mmol), **1a** (49 mg, 0.31 mmol), (*E*)-(2-iodovinyl)benzene (86 mg, 0.37 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 78 mg of **2aq** (83%) as a solid (petroleum ether/ethyl acetate=4:1): mp 89–91 °C (petroleum ether/ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.42–7.20 (m, 10H, ArH), 6.73 (d, J=16.8 Hz, 1H, CH=C), 6.36 (d, J=17.1 Hz, 1H, C=CH), 5.48 (s, 1H, one proton of C=CH_2), 5.35 (s, 1H, one proton of C=CH_2), 5.29 (t, J=7.8 Hz, 1H, OCH), 4.52 (d, J=14.7 Hz, 1H, one proton of CH_2 in Bn), 4.39 (d, J=15.0 Hz, 1H, one proton of CH_2 in Bn), 3.75 (t, J=8.9 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.22 (m, 1H, one proton of NCH_2 in oxazolidinone ring); ^{13}C NMR (75 MHz, $CDCl_3$) δ 157.7, 142.9, 136.3, 135.5, 129.3, 128.8, 128.7, 128.15, 128.06, 128.0, 126.4, 115.6, 72.0, 49.8, 48.2; MS (m/z) 305 (M^+, 4.52), 261 (M^+-CO_2, 16.52), 91 (100); IR (KBr, cm^{-1}) ν 1759, 1606, 1494, 1446, 1430, 1373, 1338, 1315, 1248, 1209, 1183, 1161, 1093, 1057. Anal. Calcd for $C_{20}H_{19}NO_2$: C 78.66, H 6.27, N 4.59. Found: C 78.55, H 6.39, N 4.58.

*4.2.18. 3-(4-Methoxybenzyl)-5-(1-phenylvinyl)-2-oxazolidinone (**2ba**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3

(83 mg, 0.60 mmol), **1b** (57 mg, 0.30 mmol), PhI (41 μL, *d*=1.82 g/mL, 74 mg, 0.36 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 82.3 mg of **2ba** (88%) as an oil (petroleum ether/ethyl acetate=4:1): ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.38–7.20 (m, 5H, ArH), 7.09 (d, *J*=8.4 Hz, 2H, ArH), 6.82 (d, *J*=8.4 Hz, 2H), 5.50 (s, 1H, one proton of C=CH_2), 5.44 (s, 1H, one proton of C=CH_2), 5.37 (t, *J*=8.0 Hz, 1H, OCH), 4.38 (d, *J*=14.7 Hz, 1H, one proton of CH_2 in Bn), 4.26 (d, *J*=14.7 Hz, 1H, one proton of CH_2 in Bn), 3.77 (s, 3H, Me), 3.53 (t, *J*=8.9 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.05 (t, *J*=7.8 Hz, 1H, one proton of NCH_2 in oxazolidinone ring); ^{13}C NMR (75 MHz, $CDCl_3$) δ 159.2, 157.6, 145.1, 137.1, 129.2, 128.6, 128.2, 127.5, 126.6, 114.2, 114.1, 73.6, 55.2, 49.0, 47.5; MS (*m*/*z*) 309 (M^+, 5.92), 265 (M^+-CO_2, 0.35), 130 (100); IR (neat, cm^{-1}) ν 1753, 1611, 1512, 1492, 1438, 1363, 1330, 1247, 1179, 1115, 1089, 1032; HRMS calcd for $C_{19}H_{19}NO_3$ (M^+): 309.1365, found 309.1364.

*4.2.19. 3-Phenyl-5-(1-phenylvinyl)-2-oxazolidinone (**2ca**).* The reaction of $Pd(PPh_3)_4$ (17.2 mg, 0.015 mmol), Cs_2CO_3 (195.4 mg, 0.60 mmol), **1c** (42.9 mg, 0.30 mmol), PhI (40 μL, *d*=1.82 g/mL, 73 mg, 0.36 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 43 mg of **2ca** (55%) as an oil (petroleum ether/ethyl acetate=10:1): ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.47 (d, *J*=8.1 Hz, 2H, ArH), 7.44–7.27 (m, 7H, ArH), 7.11 (t, *J*=7.2 Hz, 1H, ArH), 5.67–5.50 (m, 3H, C=CH_2 and OCH), 4.16 (t, *J*=8.9 Hz, 1H, one proton of NCH_2), 3.68 (dd, J_1=8.6 Hz, J_2=7.4 Hz, 1H, one proton of NCH_2); ^{13}C NMR (75 MHz, $CDCl_3$) δ 154.5, 144.7, 138.1, 137.0, 129.0, 128.9, 128.5, 126.6, 124.1, 118.3, 114.6, 73.0, 50.4; MS (*m*/*z*) 266 ($M+H^+$, 11.49), 265 (M^+, 57.91), 221 (M^+-CO_2, 49.91), 220 (100); IR (neat, cm^{-1}) ν 1755, 1599, 1501, 1404, 1307, 1223, 1138, 1018; HRMS calcd for $C_{17}H_{15}NO_2$ (M^+): 265.1103, found 265.1101.

*4.2.20. 3-Phenyl-5-(1-phenylvinyl)-2-oxazolidinone (**2ca**).* The reaction of $Pd(PPh_3)_4$ (17.1 mg, 0.015 mmol), K_2CO_3 (83.5 mg, 0.60 mmol), **1c** (46.0 mg, 0.32 mmol), PhI (40 μL, d=1.82 g/mL, 73 mg, 0.36 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 33 mg of **2ca** (39%) as an oil (petroleum ether/ethyl acetate=10:1): ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.48 (d, *J*=7.8 Hz, 2H, ArH), 7.44–7.27 (m, 7H, ArH), 7.11 (t, *J*=7.4 Hz, 1H, ArH), 5.67–5.50 (m, 3H, C=CH_2 and OCH), 4.17 (t, *J*=8.9 Hz, 1H, one proton of NCH_2), 3.69 (dd, J_1=8.7 Hz, J_2=7.2 Hz, 1H, one proton of NCH_2).

*4.2.21. 3-Benzyl-5-(2-methyl-1-phenylprop-1-en-1-yl)-2-oxazolidinone (**2da**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (84 mg, 0.60 mmol), **1d** (56 mg, 0.30 mmol), PhI (40 μL, d=1.82 g/mL, 0.36 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 70 mg of **2da** (76%) as a solid (petroleum ether/ethyl acetate=10:1 to 6:1): mp 85–86 °C (petroleum ether/ethyl ether); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.37–7.27 (m, 3H, ArH), 7.23–7.16 (m, 3H, ArH), 7.12–7.01 (m, 2H, ArH), 6.89–6.78 (m, 2H, ArH), 5.67 (dd, J_1=9.2 Hz, J_2=7.1 Hz, 1H, OCH), 4.21 (d, *J*=15.0 Hz, 1H, one proton of CH_2 in Bn), 4.13 (d, *J*=15.0 Hz, 1H, one proton of CH_2 in Bn), 3.38 (t, *J*=9.0 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.03 (t, *J*=7.8 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 1.91 (s, 3H, Me), 1.52 (s, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$) δ 158.1, 137.2, 137.1, 135.5, 131.3, 130.3, 128.5, 128.3, 127.5, 127.4, 127.0, 71.4, 47.9, 47.3, 22.9, 19.6; MS (*m*/*z*) 307 (M^+, 0.51), 263 (M^+-CO_2, 2.93), 248 (100); IR (KBr, cm^{-1}) ν 1747, 1695, 1652, 1455, 1431, 1257, 1098, 1002. Anal. Calcd for $C_{20}H_{21}NO_2$: C 78.15, H 6.89, N 4.56. Found: C 78.18, H 7.16, N 4.49.

*4.2.22. 3-Benzyl-5-(cyclohexylidene(phenyl)methyl)-2-oxazolidinone (**2ea**).* The reaction of $Pd(PPh_3)_4$ (17 mg, 0.015 mmol), K_2CO_3 (83 mg, 0.60 mmol), **1e** (68 mg, 0.30 mmol), PhI (40 μL, d=1.82 g/mL, 73 mg, 0.36 mmol), and a balloon of carbon dioxide (about 1 L) in 2 mL of DMSO afforded 77 mg of **2ea** (74%) as a solid (petroleum ether/ethyl acetate=10:1): mp 120–121 °C (petroleum ether/ethyl ether); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.40–7.00 (m, 7H, ArH), 7.00–6.70 (m, 3H, ArH), 5.72 (t, *J*=8.1 Hz, 1H, OCH), 4.20 (d, *J*=15.0 Hz, 1H, one proton of CH_2 in Bn), 4.14 (d, *J*=15.3 Hz, 1H, one proton of CH_2 in Bn), 3.37 (t, *J*=8.9 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.04 (t, *J*=7.7 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 2.50–2.20 (m, 2H, H in cyclohexylidene), 2.00–1.80 (m, 2H, H in cyclohexylidene), 1.77–1.30 (m, 6H, H in cyclohexylidene); ^{13}C NMR (75 MHz, $CDCl_3$) δ 158.1, 145.3, 137.0, 135.4, 130.3, 128.5, 128.1, 127.44, 127.36, 126.9, 70.6, 47.9, 47.4, 32.9, 30.1, 28.4, 28.1, 26.5; MS (*m*/*z*) 347 (M^+, 23.36), 303 (M^+-CO_2, 35.21), 91 (100); IR (KBr, cm^{-1}) ν 1751, 1643, 1489, 1440, 1362, 1334, 1328, 1313, 1266, 1209, 1192, 1102, 1081, 1070, 1053, 1027, 1003. Anal. Calcd for $C_{23}H_{25}NO_2$: C 79.51, H 7.25, N 4.03. Found: C 79.71, H 7.24, N 4.07.

*4.2.23. Gram scale synthesis of 3-benzyl-5-(1-phenylvinyl)-2-oxazolidinone (**2aa**).* To an oven-dried Schlenk flask were added $Pd(PPh_3)_4$ (374.9 mg, 0.324 mmol), K_2CO_3 (1.7961 g, 13.0 mmol), DMSO (10 mL), 2,3-allenyl amine **1a** (48 mg, 0.3 mmol), DMSO (5 mL), PhI (1.5922 g, 7.80 mmol), and 25 mL of DMSO under nitrogen atmosphere. The mixture was then frozen in a liquid nitrogen bath. After degassed through one freeze–thaw cycle, the argon inside was completely replaced with CO_2 by using a CO_2 balloon (about 1.5 L). After being stirred at 70 °C (temperature of oil bath) for 5 h the reaction was complete as monitored by TLC, the resulting mixture was diluted with water (40 mL) and extracted with Et_2O (25 mL×4). The organic layer was washed with brine, dried over $MgSO_4$, filtrated, and concentrated. The crude product was purified by column chromatography on silica gel (petroleum ether/ethyl acetate=4:1) to afford 1.7124 g of **2aa** (94%) as a solid: ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.35–7.22 (m, 8H, ArH), 7.20–7.10 (m, 2H, ArH), 5.52 (d, *J*=0.9 Hz, 1H, one proton of C=CH_2), 5.45 (s, 1H, one proton of C=CH_2), 5.39 (dd, J_1=8.4 Hz, J_2=7.5 Hz, 1H, OCH), 4.45 (d, *J*=15.0 Hz, 1H, one proton of CH_2 in Bn), 4.33 (d, *J*=15.0 Hz, 1H, one proton of CH_2 in Bn), 3.55 (t, *J*=8.9 Hz, 1H, one proton of NCH_2 in oxazolidinone ring), 3.07 (dd, J_1=8.7 Hz, J_2=6.9 Hz, 1H, one proton of NCH_2 in oxazolidinone ring).

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Supplementary data

Spectroscopic data, general procedure, and ^{1}H/^{13}C NMR spectra of all the products. This material is available free of charge via the internet. Supplementary data associated with this article can be found in the online version, at http://dx.doi.org/10.1016/j.tet.2013.09.087.

References and notes

1. For recent reviews, see: (a) Louie, J. *Curr. Org. Chem.* **2005**, *9*, 605; (b) Sakakura, T.; Choi, J.-C.; Yasuda, H. *Chem. Rev.* **2007**, *107*, 2365; (c) Aresta, M.; Dibenedetto, A. *Dalton Trans.* **2007**, 2975; (d) Riduan, S. N.; Zhang, Y. *Dalton Trans.* **2010**, *39*, 3347; (e) Huang, K.; Sun, C.; Shi, Z. *Chem. Soc. Rev.* **2011**, *40*, 2435; (f) Cokoja, M.; Bruckmeier, C.; Rieger, B.; Herrmann, W. A.; Kühn, F. E. *Angew. Chem., Int. Ed.* **2011**, *50*, 8510.
2. For reviews, see: (a) Dell'Amico, D. B.; Calderazzo, F.; Labella, L.; Marchetti, F.; Pampaloni, G. *Chem. Rev.* **2003**, *103*, 3857; (b) Chaturvedi, D.; Ray, S. *Curr. Org. Chem.* **2007**, *11*, 987.
3. For reports on the synthesis of oxazolidinone from aziridines and CO_2, see: (a) Kawanami, H.; Matsumoto, H.; Ikushima, Y. *Chem. Lett.* **2005**, *34*, 60; (b) Miller, A. W.; Nguyen, S. T. *Org. Lett.* **2004**, *6*, 2301; (c) Sudo, A.; Morioka, Y.; Sanda, F.; Endo, T. *Tetrahedron Lett.* **2004**, *45*, 1363; (d) Sudo, A.; Morioka, Y.; Koizumi, E.; Sanda, F.; Endo, T. *Tetrahedron Lett.* **2003**, *44*, 7889; (e) Shi, M.; Jiang, J.; Shen, Y.;

Feng, Y.; Lei, G. *J. Org. Chem.* **2000**, *65*, 3443; (f) Du, Y.; Wu, Y.; Liu, A.; He, L. *J. Org. Chem.* **2008**, *73*, 4709; (g) Hancock, M. T.; Pinhas, A. R. *Synthesis* **2004**, *14*, 2347; (h) Mu, W.-H.; Chasse, G. A.; Fang, D.-C. *J. Phys. Chem. A* **2008**, *112*, 6708; (i) Wu, Y.; Liu, G. *Tetrahedron Lett.* **2011**, *52*, 6450.

4. For reports on the reaction of 1,2-aminoalcohols with CO_2 (60 bar) producing oxazolidinone with low conversion (10–97%) see: Bhanage, B. M.; Fujita, S.; Ikushima, Y.; Arai, M. *Green Chem.* **2003**, *5*, 340.
5. For reports on the synthesis of oxazolidinones from propargylalcohols, amines, and CO_2 (2.5 MPa or supercritical conditions), see: (a) Gu, Y.; Zhang, Q.; Duan, Z.; Zhang, J.; Zhang, S.; Deng, Y. *J. Org. Chem.* **2005**, *70*, 7376; (b) Jiang, H.; Zhao, J.; Wang, A. *Synthesis* **2008**, 763.
6. For reports on the synthesis of oxazolidinones from propargylamines and CO_2 (5–40 bar), see: (a) Costa, M.; Chiusoli, G. P.; Rizzardi, M. *Chem. Commun.* **1996**, 1699; (b) Shi, M.; Shen, Y. *J. Org. Chem.* **2002**, *67*, 16; For the synthesis of oxazolidinones from propargylamines, which were in situ generated from terminal alkynes, aldehyde, amines, and CO_2 (1 atm) see: (c) Yoo, W.; Li, C. *Adv. Synth. Catal.* **2008**, *350*, 1503.
7. For reports on the $Pd(OAc)_2$ catalyzed reaction of 2,3-allenyl amines (substrates scope was limited in single substituent group in allene moiety.) with CO_2 (11.5 MPa) producing 5-vinyloxazolidin-2-ones in very low to moderate yields (2–65%) see: Kayaki, Y.; Mori, N.; Ikariya, T. *Tetrahedron Lett.* **2009**, *50*, 6491.
8. For reports on the reaction of amides with CO_2, see: (a) Chen, G.; Fu, C.; Ma, S. *Org. Lett.* **2009**, *11*, 2900; (b) Chen, G.; Fu, C.; Ma, S. *Org. Biomol. Chem.* **2011**, *9*, 105.
9. For comprehensive reviews, see: (a) Zappia, G.; Gacs-Baitz, E.; Delle Monache, G.; Misiti, D.; Nevola, L.; Botta, B. *Curr. Org. Synth.* **2007**, *4*, 81; (b) Zappia, G.; Cancelliere, G.; Gacs-Baitz, E.; Delle Monache, G.; Misiti, D.; Nevola, L.; Botta, B. *Curr. Org. Synth.* **2007**, *4*, 238; (c) Evans, D.; Takacs, J.; McGee, L.; Ennis, M.; Mathre, D.; Bartroli, J. *Pure Appl. Chem.* **1981**, *53*, 1109; (d) Evans, D. *Aldrichimica Acta* **1982**, *15*, 23; (e) Ager, D. J.; Prakash, I.; Schaad, D. R. *Chem. Rev.* **1996**, *96*, 835.
10. Crystal data for compound **2ab**: $C_{19}H_{19}NO_2$, MW=293.35, monoclinic, space group $P2(1)/c$, final R indices [$I>2\sigma(I)$], $R1$=0.0477, $wR2$=0.1253; R indices (all data), $R1$=0.0584, $wR2$=0.1341; a=15.4736(11) Å, b=5.9200(4) Å, c=17.4387(12) Å, α=90°, β=107.893(2)°, γ=90°, V=1520.18(18) Å^3, T=173(2) K, Z=4, reflections collected/unique 16,901/2664 (R_{int}=0.1088), number of observations [$>2\sigma(I)$] 2157, parameters: 207. Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Centre. CCDC 948729.
11. Ye, J.; Li, S.; Ma, S. *Org. Biomol. Chem.* **2013**, *11*, 5370.